

CHE331: Petroleum Refinery Engineering
(Credit: 03)

Unit wise Course Schedule:

Unit-1

History and development of refining; composition of crude oil; evaluation of oil stocks
(6 Hour)

Unit-2

Refinery and distillation processes; ADU, VDU, auxiliary process and operations; refinery products and test methods **(9 Hour)**

Unit-3

Physical properties of petroleum oil; light, middle and heavy distillates; lubricating oil **(6 Hour)**

Unit-4

Chemical treatment; extraction process; dewaxing. Visbreaking, reforming, coking, polymerization, alkylation and dehydrogenation processes. **(9 Hour)**

Unit-5

Natural and refinery gases, recent advances on refinery operations; environmental impacts
(3 Hour)

Unit-6

Designing of refinery equipment **(6 Hour)**

Books:

1. Nelson, W.L., "Petroleum Refinery Engineering" McGraw-Hill Book Co.
2. Gary, J.H., Handwerk, G.E., "Petroleum Refining: Technology & Economics", Marcel Dekker, Inc.
3. NPTEL courses (suggested)